SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) **Department of Computer Science and Engineering**

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	**divyasree.p**	**Reg. No.:**	**192565016**
Section / Batch:	**BE CSE cyber security**	**Date:**	**27-07-2026**

Instructions to Students

• Answer the following question. Total Marks: 100.

• Carefully analyze the given scenario and identify the computational problem involved.

• Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.

• Clearly justify the chosen solution approach with proper reasoning and analytical interpretation. • Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach. • Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.

• Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

53) university system stores attendance records unsorted due to frequent updates. Faculty need to quickly locate specific student records.

a) Explain how Linear Search works in this scenario.

b) Analyze its performance in best, average, and worst cases.

c) Justify whether sorting would improve efficiency.

Code of Conduct

I kaviya R(192521075) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kaviya R

RUBRICS FOR CALCULATION

Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	**Course Coordinator**	**Head of Department**
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem

A university stores attendance records in an **unsorted** list because records are updated frequently.

Attendance Records (Unsorted)

Position	**Student ID**
1	S101
2	S105
3	S103
4	S108
5	S102
6	S110
7	S106

Target Student: S110

a. **Execution of Linear Search**

Step 1: Compare S110 with S101 → Not Found ❌

Step 2: Compare S110 with S105 → Not Found ❌

Step 3: Compare S110 with S103 → Not Found ❌

Step 4: Compare S110 with S108 → Not Found ❌

Step 5: Compare S110 with S102 → Not Found ❌

Step 6: Compare S110 with S110 → Found ✅

Student S110 is located at Position 6.

Number of Comparisons = 6

Execution Table

Step	**Record Checked**	**Comparison**	**Result**
1	S101	S101 = S110?	No
2	S105	S105 = S110?	No
3	S103	S103 = S110?	No
4	S108	S108 = S110?	No
5	S102	S102 = S110?	No
6	S110	S110 = S110?	**Found**

a. **Performance Analysis**

Best Case

- Target is at the **first position**.
- Comparisons = **1**
- **Time Complexity = O(1)**

Average Case

- Target is somewhere in the **middle**.
- Comparisons ≈ **n/2**
- **Time Complexity = O(n)**

Worst Case

- Target is at the **last position** or not present.
- Comparisons = **n**
- **Time Complexity = O(n)**

Performance Table

Case	**Comparisons**	**Time Complexity**
Best	1	O(1)
Average	n/2	O(n)
Worst	n	O(n)

a. **Would Sorting Improve Efficiency?**

Yes, if searches are performed frequently.

- After sorting the records, **Binary Search** can be used.
- Binary Search has **O(log n)** search time, which is much faster than Linear Search.
- However, because attendance records are **updated frequently**, sorting after every update takes extra time.
- Therefore, **keeping the records unsorted and using Linear Search is more practical** for this university system.

Final Conclusion

- Linear Search checks each attendance record **one by one**.
- In this execution, the student **S110** was found at **Position 6** after **6 comparisons**.
- Best Case = **O(1)**, Average Case = **O(n)**, Worst Case = **O(n)**.
- Sorting improves search speed only when records change infrequently. Since attendance records are updated often, **Linear Search is the appropriate choice**.